SUPPLY CHAIN ANALYTICS

Python Visualizations, Business Questions & Insights

DataCo Global Supply Chain Dataset | Portfolio Appendix

Project context

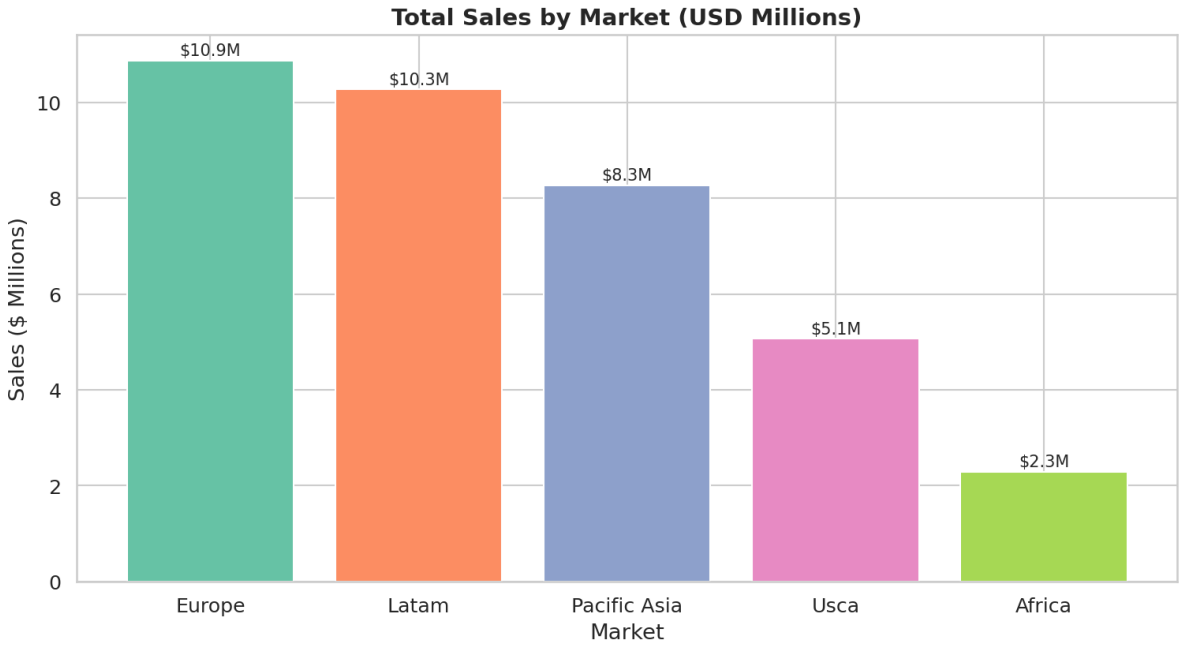
This document extracts the chart visuals embedded in the supplied Supply Chain Analytics project and pairs them with the business question, reported interpretation, business impact, and recommended action. The source describes 180,519 records, 65,752 unique orders, 20,652 customers, and data spanning 2015–2018.

Interpretation note: the content below is grounded in the supplied project document. Where its narrative and plotted chart appear inconsistent, that discrepancy is called out rather than silently reconciled. Validate calculations against the underlying dataset before presenting any metric as definitive.

Charts and business interpretation

Chart 1 — Total Sales by Market

Business question: Which geographic market generates the most revenue?



Business insight: LATAM leads with total sales of about $9.5M, followed closely by Europe. Africa is the smallest market at approximately $2.1M, suggesting a potential expansion opportunity.

Business impact: Revenue contribution should guide market-level marketing and logistics investment.

Recommended action: Protect the LATAM and Europe core markets; assess Africa's demand, infrastructure, and fulfillment economics before expanding.

Chart 2 — Total Profit by Market

Business question: Which market is most profitable, and does it align with sales rank?



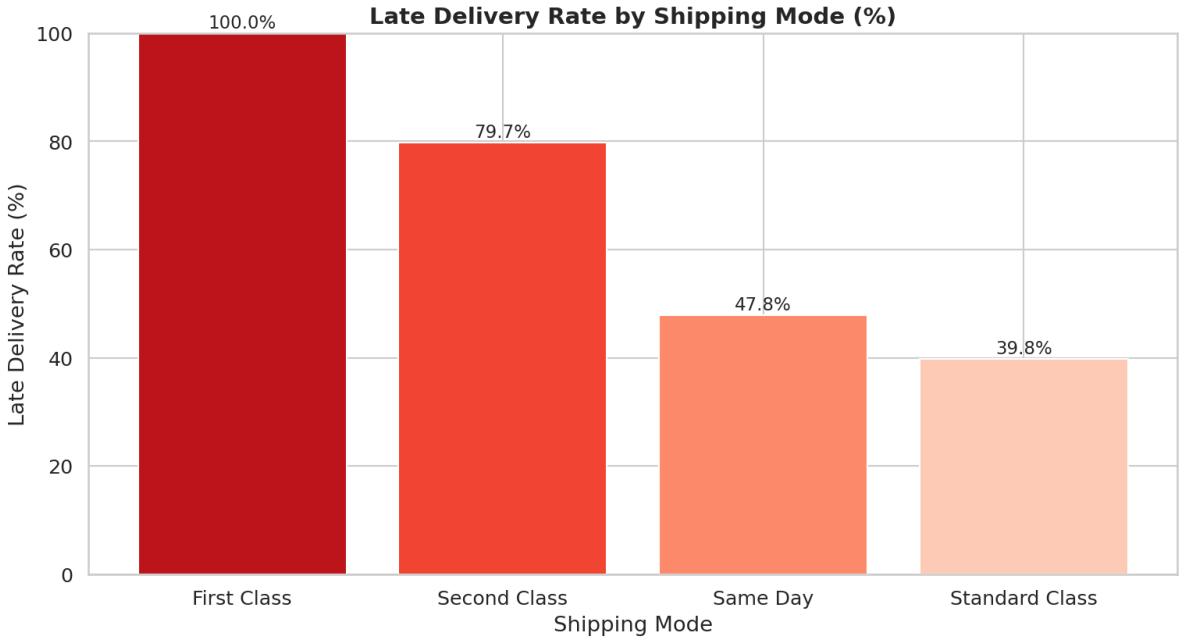
Business insight: Europe generates the highest profit despite being second in sales. LATAM's profit does not match its revenue rank, potentially indicating higher costs or deeper discounting.

Business impact: Profit-to-revenue comparison helps distinguish scale from operating efficiency.

Recommended action: Benchmark Europe's operating model and investigate LATAM's logistics costs, product mix, and discounting.

Chart 3 — Late Delivery Rate by Shipping Mode

Business question: Which shipping mode has the worst on-time delivery performance?



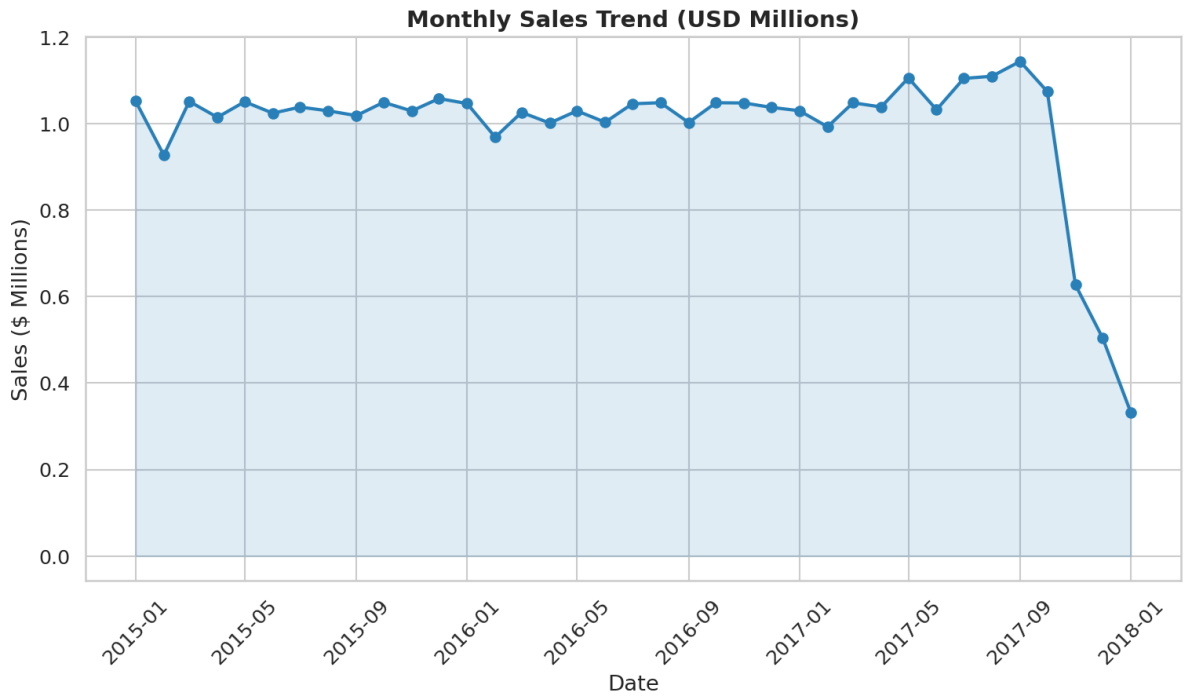
Business insight: The report identifies Same Day shipping as having the highest late-delivery rate in its narrative; the displayed chart labels show First Class at 100%, Second Class at 79.7%, Same Day at 47.8%, and Standard Class at 39.8%. This is an inconsistency between the narrative and plotted values.

Business impact: Customers may receive poor service despite paying for faster shipping; inconsistent chart interpretation can lead to the wrong operational decision.

Recommended action: Validate the calculation and category mapping before acting. Then audit the shipping mode with the highest verified late-delivery rate.

Chart 4 — Monthly Sales Trend

Business question: Does revenue show seasonality or a growth trend over time?



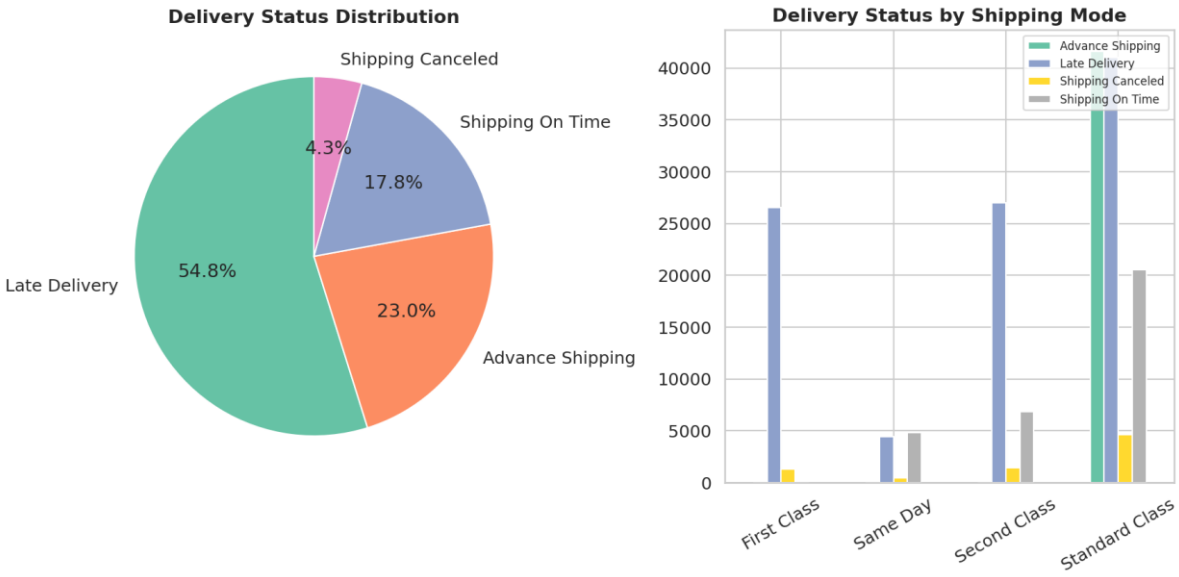
Business insight: The report describes recurring Q4 peaks and relatively stable sales outside peak periods. The plotted series drops sharply near the end of the available date range, which may reflect incomplete coverage in the final months.

Business impact: Seasonality can help plan inventory, warehouse labor, and carrier capacity, but incomplete periods can distort trends.

Recommended action: Validate the final months and compare complete periods year over year. Prepare inventory and carrier capacity ahead of the Q4 peak.

Chart 5 — Delivery Status Distribution

Business question: How are deliveries distributed across performance categories?



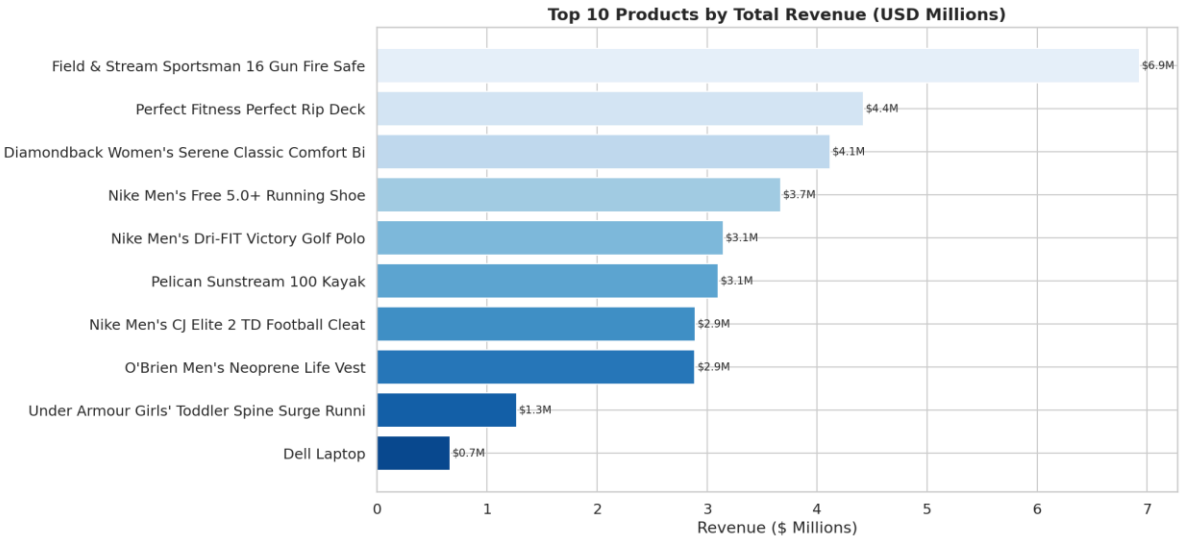
Business insight: The report states that Late Delivery accounts for 54.8%, Advance Shipping 23.0%, Shipping On Time 17.8%, and Shipping Canceled 4.3%.

Business impact: More than half of deliveries are categorized as late, indicating a major service reliability problem.

Recommended action: Track late-delivery rate weekly by region, shipping mode, and carrier; set a measurable reduction target and monitor SLA compliance.

Chart 6 — Top 10 Products by Revenue

Business question: Which specific products drive the most revenue?



Business insight: The report highlights Field & Stream Sportsman 16 Gun Safe, Perfect Fitness Perfect Rip Deck, and Diamondback Women's bicycles among the leading products.

Business impact: Revenue concentration makes availability of top-selling SKUs especially important; stockouts could disproportionately affect sales.

Recommended action: Set appropriate safety stock for high-revenue SKUs, monitor forecast error, and evaluate alternate suppliers for critical products.

Chart 7 — Customer Segment Performance

Business question: How do Consumer, Corporate, and Home Office segments compare on key metrics?

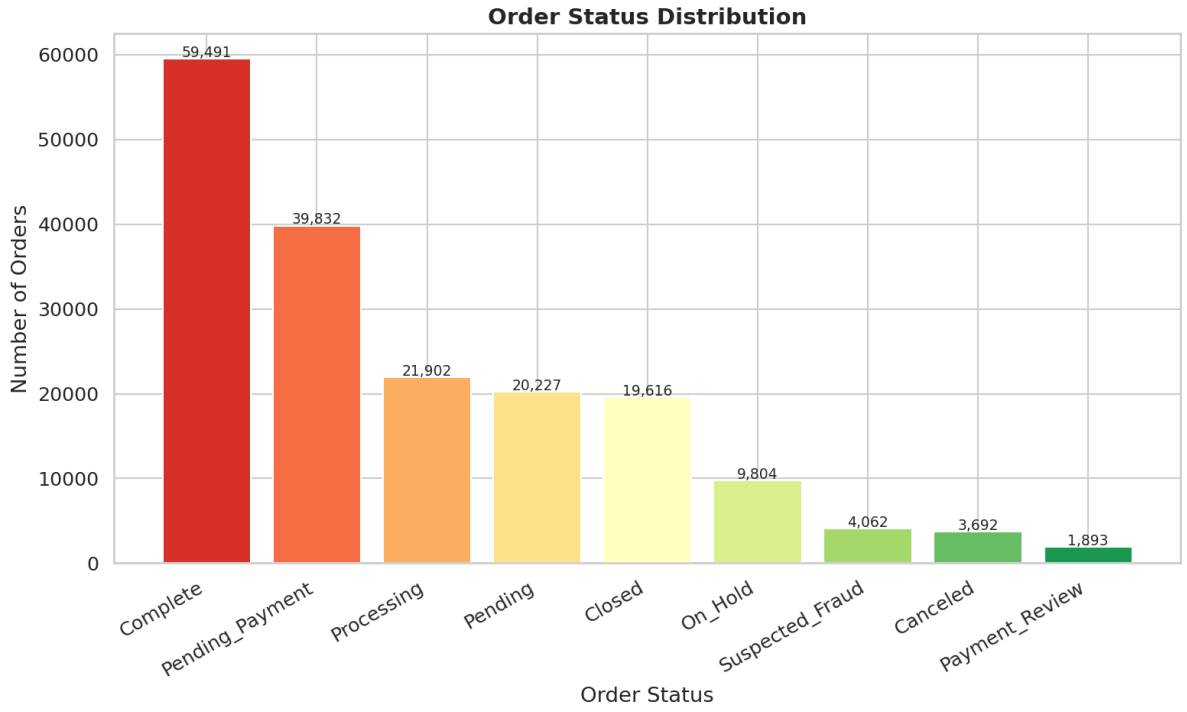
Business insight: The report says Consumer leads in volume and total contribution, Corporate has higher average profit per order, and Home Office has the highest average order value.

Business impact: Different segments have different value drivers and should not automatically receive identical commercial strategies.

Recommended action: Use volume and loyalty programs for Consumer, account management for Corporate, and tailored premium offerings for Home Office; verify the segment metrics before implementation.

Chart 8 — Order Status Distribution

Business question: What proportion of orders reach completion versus remaining pending, processing, or canceled?



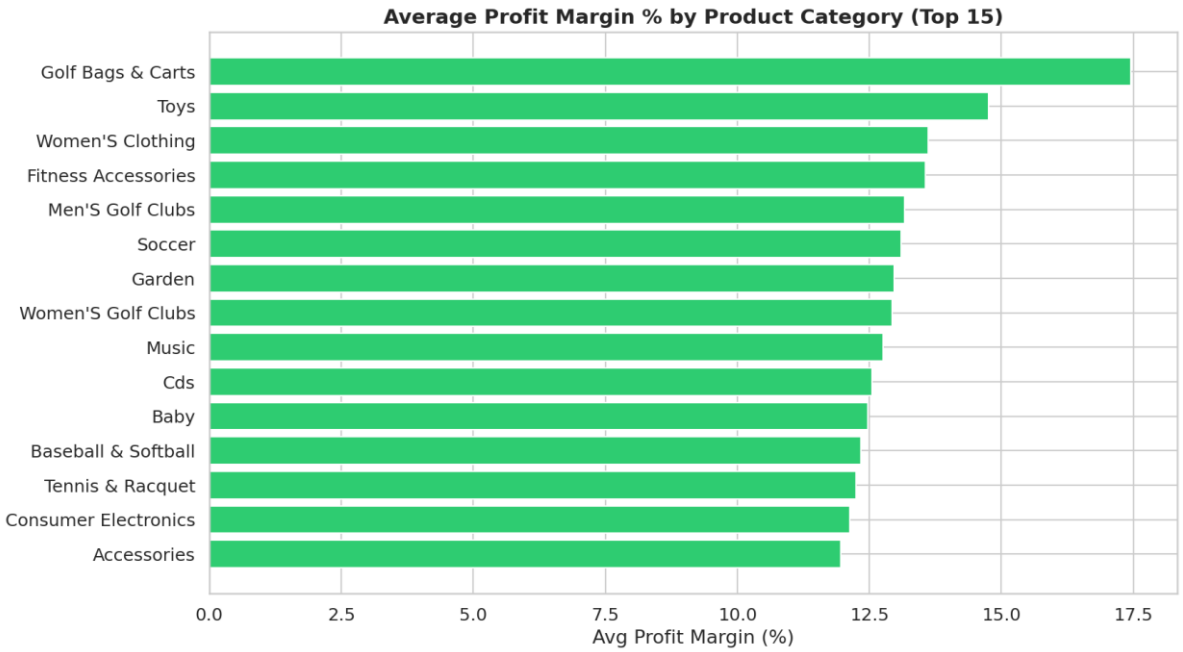
Business insight: The report states that COMPLETE orders represent about 33% of records, while PENDING_PAYMENT is around 22% and PROCESSING around 12%. The chart includes several additional statuses.

Business impact: Orders that remain pending or on hold can delay fulfillment and cash collection, though status definitions should be checked before treating them as lost revenue.

Recommended action: Create aging buckets for pending and held orders, define escalation rules, and distinguish order-item rows from unique orders in the calculation.

Chart 9 — Profit Margin by Product Category

Business question: Which product categories are most and least profitable?



Business insight: The report notes substantial variation in average profit margin by category, with some categories potentially having negative margins. The plotted chart shows the top 15 categories by average profit margin.

Business impact: Category-level averages can hide unprofitable products and discount-driven margin erosion.

Recommended action: Drill down to SKU-level profitability, then review pricing, supplier costs, and discounts for categories or products with persistently low margins.

Chart 10 — Discount Rate vs. Profit Relationship

Business question: Does offering higher discounts reduce profit?



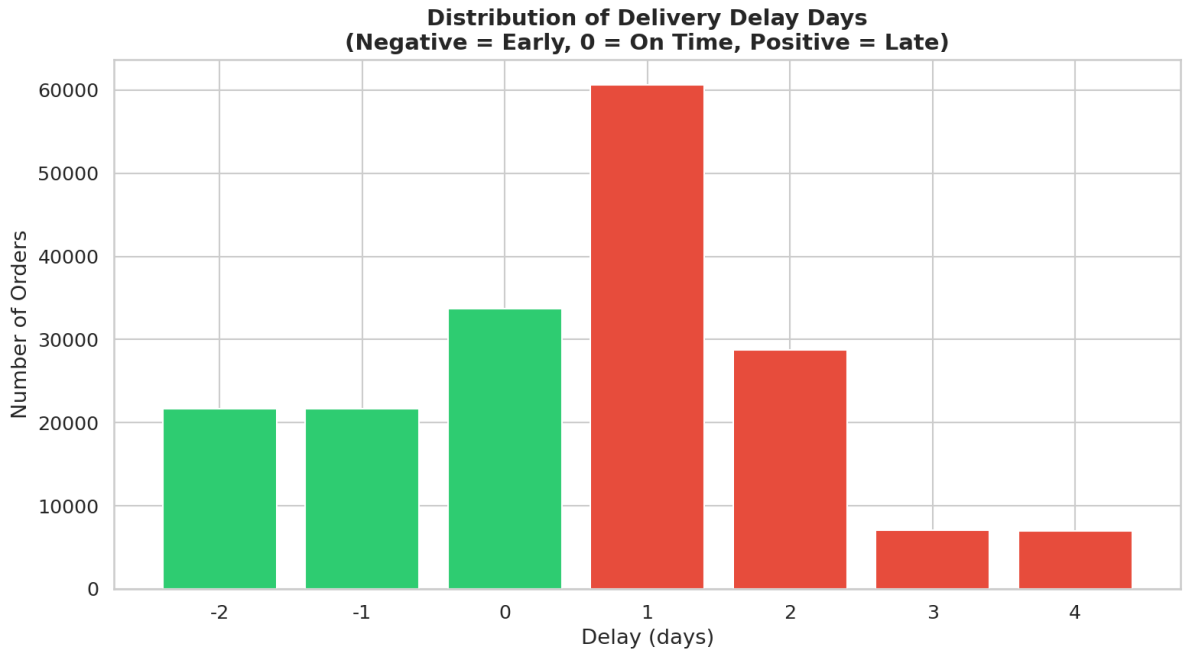
Business insight: The report describes a negative relationship between discount rate and profit and states that orders with discounts above 30% are almost universally negative-profit. The scatter plot is broad, so this should be validated with grouped summaries.

Business impact: Uncontrolled discounting may erode margin even when it supports sales volume.

Recommended action: Calculate profit and margin by discount band while controlling for product/category mix. Introduce approval thresholds only after validating the relationship.

Chart 11 — Delivery Delay Distribution

Business question: What is the distribution of delivery delays in days?



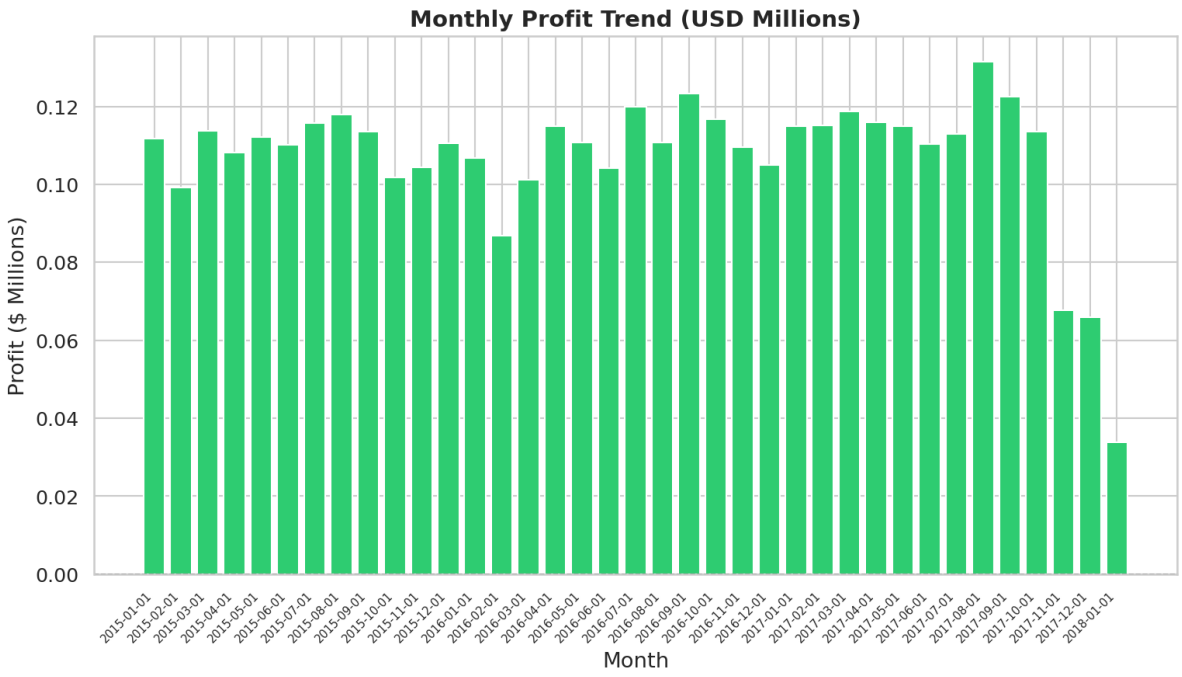
Business insight: The report identifies a delay of +3 days as the single most common outcome. The plotted chart labels and bars should be checked against the report's stated feature definition before finalizing exact counts.

Business impact: A consistent delay pattern can indicate that promised delivery dates or fulfillment planning are not aligned with actual performance.

Recommended action: Measure delay as actual shipping days minus scheduled shipping days, validate signs and units, then adjust ETA promises and proactively notify customers.

Chart 12 — Monthly Profit Trend

Business question: Is profitability consistent month over month or volatile?



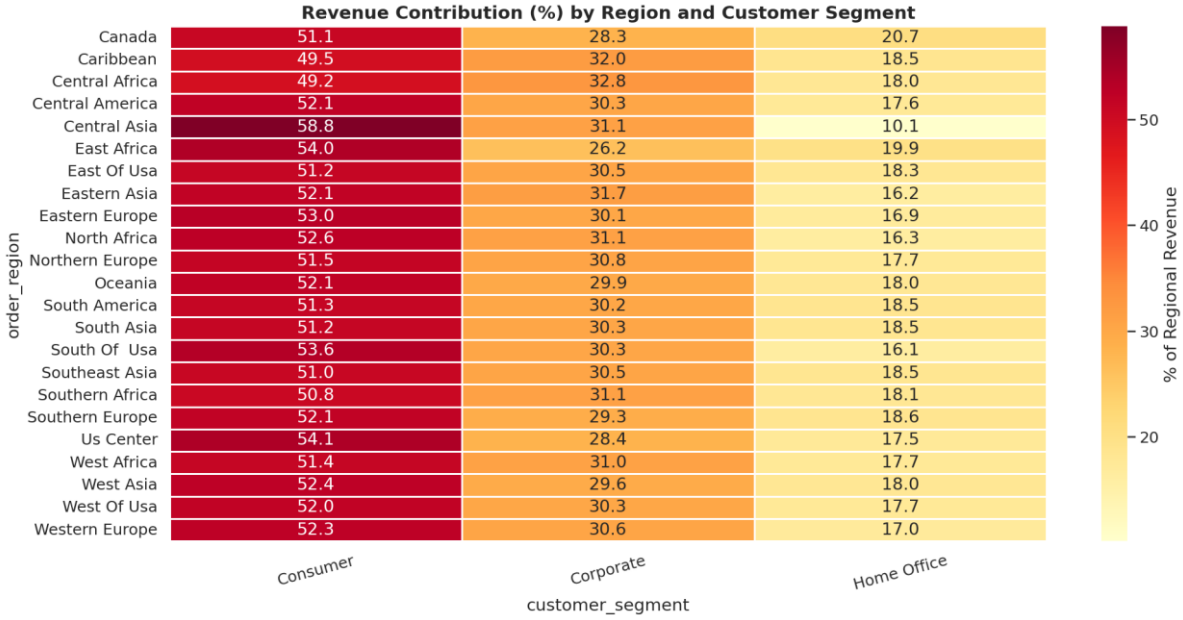
Business insight: The report describes month-to-month profit volatility and mentions some negative-profit months, especially around Q1 and promotional periods. The chart included in the source file is not clearly distinguishable from other chart images, so no exact monthly values are reproduced here.

Business impact: Volatility complicates financial planning and may reflect seasonality, product mix, costs, or promotions.

Recommended action: Build a monthly profit bridge by revenue, discount, product mix, and shipping cost; investigate negative months before setting profit floors.

Chart 13 — Region × Segment Revenue Heatmap

Business question: How does customer-segment revenue contribution vary by geographic region?



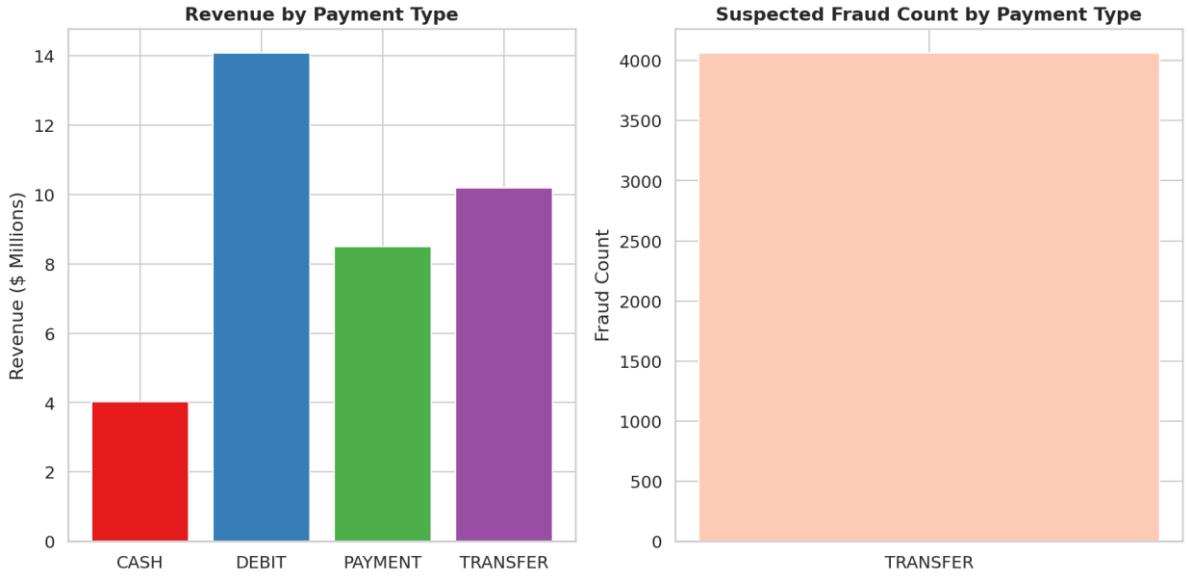
Business insight: The report says Consumer dominates in most regions, while some regions have a relatively stronger Corporate contribution.

Business impact: The heatmap can identify where B2B sales efforts may have the greatest opportunity.

Recommended action: Prioritize corporate account development in regions with meaningful Corporate contribution, after checking regional sample sizes and revenue shares.

Chart 14 — Payment Type Analysis

Business question: Which payment type is associated with the most suspected fraud?



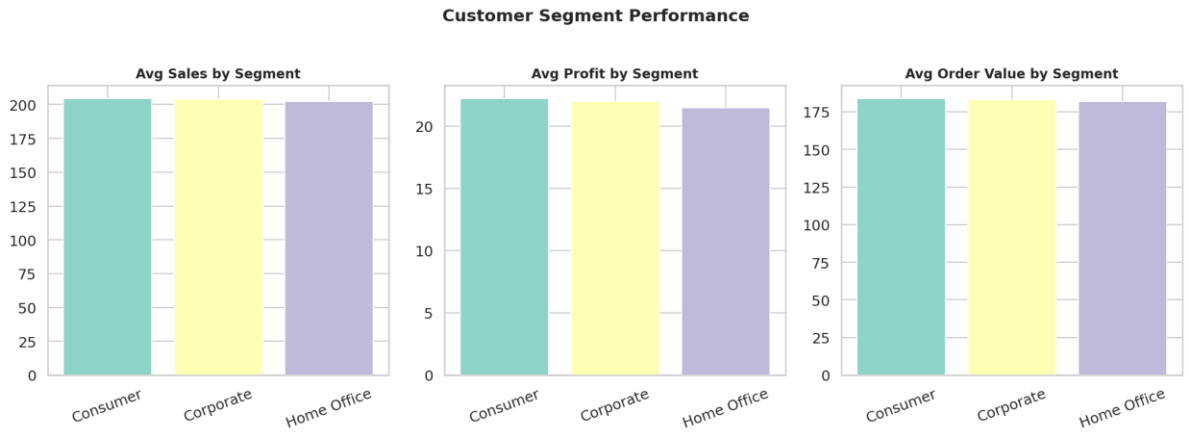
Business insight: The report says debit is the most common payment type and recommends comparing fraud patterns across payment methods. The source chart's fraud panel appears to show only Transfer, so the underlying grouping should be validated.

Business impact: Fraud controls should be based on fraud rate and loss exposure, not only the raw count of transactions by payment type.

Recommended action: Calculate suspected-fraud rate and value by payment type using a consistent denominator; strengthen verification for channels with demonstrably higher risk.

Chart 15 — Quarterly Performance by Year

Business question: Does the Q4 sales spike repeat consistently every year?



Business insight: The report describes Q4 as a recurring strong quarter for sales and profit across the available years. The final year appears incomplete, so comparisons for that year may be misleading.

Business impact: Seasonal patterns can inform capacity and inventory planning when complete quarters are compared consistently.

Recommended action: Compare like-for-like completed quarters and begin Q4 readiness planning several weeks ahead of the peak.

Cross-chart takeaways

- Delivery reliability is the clearest operational concern in the source report; validate the late-delivery definition and segment results by market, shipping mode, and carrier.
- Profitability needs to be analyzed alongside revenue. Market and category averages can conceal loss-making products, discount effects, and differences in cost-to-serve.
- Recurring seasonality can support inventory and logistics capacity planning, but incomplete periods should not be compared with full periods.
- Data quality and metric governance matter: resolve inconsistent chart narratives, verify denominators (rows versus unique orders), and reproduce KPI calculations from the cleaned dataset before sharing externally.